

Chapter 5

Proposed Work

5.1 Group Parameter Evaluation

It is vital to evaluate the DNP group parameters to ensure they are accurate and usable. To assess accuracy, the parameters are validated using macroscopic experimental data and MSRE transients. To evaluate usability, modified versions of the MSRE and MSFR are analyzed using the spatially resolved flow model. These evaluations demonstrate the validity and practical applications of the DNP group parameters.

5.1.1 Validation

Validating the DNP group parameters is essential to ensure their practical utility. Although the group parameters are verified with other group parameters in the literature, this does not necessarily mean they capture all physical phenomena of interest. Validation identifies potential weaknesses in the approach by highlighting discrepancies between predicted and experimental results.

Validating the group parameters generated in this work is conducted in two stages, as discussed in Section 3.5. The first stage validates the delayed neutron count rate formulation and group generation for a static problem. The second stage validates the updated DNP group parameters by comparing their reactivity feedback with that of the static group parameters.

The first stage validates the irradiation by comparing the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [76]. This requires modeling Godiva in OpenMC and solving the k-eigenvalue problem to irradiate the sample. The post-irradiation sample concentration is passed to the scripts that generate the DNP group parameters. The delayed neutron count rate and group parameters are compared with values from Keepin et al. to identify discrepancies. Some differences are expected due to uncertainties in the data and experimental results, so validation is successful if the results align within a few percent. If they do not align, this indicates either an issue with the scripts or with the data. If the issue lies with the data, it will be noted for future work to improve the microscopic data, enabling a more successful implementation of the proposed methodology.

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indicates either a bug not present during verification or an issue with the data. If the issue lies with the data, it will be noted for future work to improve the microscopic data, enabling more successful implementation of the proposed methodology.

The second stage validates the generated DNP group parameters. This stage uses MSRE pump start-up and coast-down experimental data to assess the accuracy of the group parameters in modeling reactivity changes from DNPs [126]. This stage requires modeling the MSRE core and upper and lower plena in Moltres [51]. The reactivity effects of the DNPs are calculated for both transients using static and updated DNP group parameters. The updated group parameters are expected to predict reactivity more accurately as they account for more physical phenomena in the MSRE. If neither set of reactivity responses aligns with the experimental results, then there is a model issue, as Moltres has previously demonstrated the ability to model these transients [129, 116]. These transients have also been evaluated using other tools, providing additional comparison [46, 37]. If the static group parameters predict reactivity more accurately than the update group parameters, this suggests an issue with the updated group parameter formulation, particularly if the first validation stage succeeded. The issue must be resolved before proceeding, as the updated DNP group parameters should provide higher accuracy in MSR modeling.

5.1.2 Reactor Generalization

One of the desired outcomes for this work is formulation of DNP group parameters for MSRs which can be applied generally. If the DNP group parameters can be applied generally to different systems with similar residence times, then the amount of computational work required is drastically reduced. However, it is necessary that sufficient evidence is provided demonstrating that the generated parameters are identical for different MSRs.

Specifically, the MSRE and the MSFR will be simulated using the spatially resolved model [134, 24]. The models will be adjusted such that the in-core and ex-core residence times are identical for each. The result will be a set of group parameters for two distinct reactors but with the same residence time tabulations. Instead of calculating the neutron flux, both reactors will be given an incident neutron energy spectra for irradiating the sample. The impact of the incident neutron energy spectra is discussed in Section 5.2.4, so differences caused by the spectra will not be studied in this section. Additionally, the fuel salt and fuel temperature in the MSFR will be replaced with those from the MSRE.

The two different reactor simulations will have all of the same input parameters aside from the mesh and thermal-hydraulic parameters. If the DNP group parameters have negligible differences, then the geometry and thermal-hydraulic parameters are not necessary when generating the group parameters. Instead, all that is needed are the other input parameters, meaning the work does not need to be replicated for every different reactor geometry and thermal-hydraulic configuration. If the DNP group parameters have non-negligible differences, then each MSR needs to have DNP group parameters generated for its own specific geometry and flow conditions. Further investigation may also be warranted to determine if there are one or two parameters which could be used to capture these differences, thus reducing the total amount of work necessary to model all MSR designs.

5.2 Data Evaluation

The use and handling of various data directly impact the DNP group parameters. These impacts arise from the data source, time resolution of fission yields, number of DNPs included, and incident neutron energy used to irradiate the sample. The data source includes cross-section data, fission yields, emission probabilities, half-lives, and other associated data used to generate the DNP group parameters. The time resolution of fission yields refers to the difference between using cumulative fission yields and time-resolved depletion chains to evaluate DNP

concentrations. The number of DNPs is critical, as modeling more DNPs increases computational cost for the spatially resolved model, so minimizing this term without affecting group generation is desirable. Finally, the incident neutron energy spectra will be investigated to determine the most effective way to organize the DNP group parameters by energy.

5.2.1 Data Sources

Huynh et al. compare the effects of cumulative fission yields and emission probabilities from Joint Evaluated Fission and Fusion (JEFF), ENDF, and Japanese Evaluated Nuclear Data Library (JENDL) on the total delayed neutron yield [69]. Mathieu et al. investigate additional datasets, including the National Nuclear Data Center (NNDC) website (which sources data from Evaluated Nuclear Structure Data File (ENSDF)) and various data compilations [101, 122, 7, 139]. These comparisons can be expanded to include the IAEA database, NNDC website, JEFF, ENDF, JENDL, ENSDF, and other data compilations for emission probabilities (delayed neutron branching ratios), half-lives, fission yields, and cross sections. Comparing these data sources clarifies the uncertainty caused by differences in the data. Table 5.1 shows the data sources used in the comparison. The "Concentration" column refers to data including fission yields and cross-sections that affect the DNP concentration term. Although half-lives also influence concentration, data sources without a concentration contribution provide only half-life and emission probability data. Audi et al., Pfeiffer and Kratz, and Rudstam et al. present data from the literature rather than official databases. Audi et al. use ENSDF, Atomic Mass Evaluation (AME), experimental data from the literature, and trends in neighboring nuclides. Pfeiffer and Kratz present experimental data, an update to the Kratz Herrmann formula (KHF), and data from a unified macroscopic-microscopic model within the quasi-particle random-phase approximation (QRPA). Rudstam et al. present experimental data.

Data name	Year	Source	Half-life	Emission probability	Concentration
IAEA database	2018	[40]	Yes	Yes	No
JEFF 3.3	2017	[125]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[25]	Yes	Yes	Yes
JENDL-5	2021	[72]	Yes	Yes	Yes
ENSDF	2025	[44]	Yes	Yes	No
Audi et al.	2003	[7]	Yes	Yes	No
Pfeiffer and Kratz	2002	[122]	Yes	Yes	No
Rudstam et al.	1993	[139]	Yes	Yes	No

Table 5.1: Various data sources as well as the data which they contain.

The expected outcome is that recently updated data and official databases will be more accurate than collections from the literature. Recent data should be more accurate due to improved equipment and measurement techniques. Official databases are expected to be more reliable, as they have many users reporting errors and have undergone extensive development over time. Investigating these data sources and comparing the DNP group parameters generated from each will be useful when analyzing uncertainty. If the differences are large, understanding the uncertainty propagation will be important. If the differences are negligible, this will simplify further uncertainty analysis.

5.2.2 Fission Yield Time Resolution

The time resolution for formation of DNPs is irrelevant for stationary, saturation irradiation DNP group parameter formulations, allowing the use of cumulative fission yields instead of full depletion simulations. However, the time resolution of DNP formation needs to be considered for group parameter calculations including circulation. This is an important consideration, as if using the cumulative fission yield generates the same group parameters as the full depletion simulation, then the computational cost would be greatly reduced. It is also useful because the spatially resolved method using the cumulative fission yield is significantly cheaper than full spatially resolved depletion.

Before comparing the methods for a circulating system, the methods are first compared using a stationary system. The DNP group parameters should align for both methods in a stationary system. This approach is common in the literature, so if the results do not align, this indicates an issue with the implementation.

To evaluate the efficacy of using cumulative fission yields compared to depletion, a direct comparison will be made between the group parameters generated by each method. The same dataset, ENDF/B-VIII.0, will be used for both to eliminate differences from data. The DNP group parameters will be generated using each method for various residence times to capture time-dependent effects at higher resolution.

If the DNP group parameters are identical for all residence times, this will significantly reduce computational costs, and the simpler method will be implemented. If non-negligible differences exist, the more costly full depletion method will continue to be used.

5.2.3 Number of DNP

The number of DNPs to track is of particular interest for the spatially resolved flow model. The 0D flow model's computational cost increases negligibly with more DNPs, while the spatially resolved flow model requires significantly more memory and operations to track each DNP over space and time. To improve computational efficiency, minimizing the number of DNPs is desirable. If the 0D flow model generates DNP group parameters with negligible differences compared to the spatially resolved flow model, minimizing the number of DNPs becomes less critical. However, it is of interest to optimize the spatially resolved flow model for potential future analyses which could include spatially resolved reprocessing and other effects.

The impact of each DNP is given in Equation (2.21), replicated here:

$$\bar{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}.$$

Dividing $\bar{v}_{d,i}$ by the total delayed neutron yield, $\bar{v}_d$, provides the relative contribution of each DNP to the total delayed neutron yield. The DNPs with the most impact can be modeled preferentially to determine the minimum number of DNPs to model with negligible changes to the DNP group parameters. A full sensitivity study will be conducted to determine a reasonable number of DNPs to include in the spatially resolved flow model.

5.2.4 Incident Neutron Energy Spectra

The incident neutron energy spectrum directly affects the DNP group parameters. However, it would be computationally expensive to generate new DNP group parameters for every unique incident neutron energy spectrum. Some coarser resolution is required, so it is of interest to determine a reasonable categorization of the DNP group parameters for different energy spectra.

This sensitivity study will investigate different levels of fidelity for organizing the energy spectra: fine, middling, coarse, and single-value spectra. The approach uses the 0D flow model, with changes only to the neutron source

energy spectra. The fine, middling, and coarse spectra include the full incident neutron energy spectra but with different bin sizes. The single-value spectra include several categories: neutron energy classification, mean energy, and median energy. The neutron energy classification is provided in Table 5.2, recreated from [98]. The study will also examine hand-crafted spectra where the mean and median remain unchanged, but the spectrum itself varies.

Category	Energy Range
Cold	$< 0.003 [eV]$
Thermal	$0.003 - 0.4 [eV]$
Epithermal	$0.4 - 100 [eV]$
Intermediate	$0.1 - 200 [keV]$
Fast	$0.2 - 10 [MeV]$
Relativistic	$> 10 [MeV]$

Table 5.2: Classification of neutron energy categories by energy range from [98].

This sensitivity study will determine the necessary level of detail in the incident neutron energy spectrum for generating DNP group parameters. Organizing by category or mean energy is expected to suffice for most cases. For cases where this approach is inadequate, a more robust method can be selectively applied.

5.2.5 Irradiation Combination

A key aspect of the methodology is combining pulse and saturation irradiation DNP group parameters. The current approach is similar to Brady's approach, combining the group 5 and 6 parameters from the pulse irradiation with the remaining parameters from the saturation irradiation to form the six group fit parameters [21]. Another approach is the Levenberg-Marquardt method, which combines pulse and saturation irradiation data [91, 100, 106]. This method of fitting DNP group parameters to both datasets simultaneously may yield more accurate results. This can also be accomplished using the trust region reflective (TRF) approach, which allows for bounds to be included for the parameters [22].

The proposed methods to compare are the current method, the Levenberg-Marquardt method, and the TRF method. The new methods minimize the residual of the form:

$$e = \min_x \left\| \frac{A_p x - b_p}{b_p} + \frac{A_s x - b_s}{b_s} \right\|_2^2, \quad (5.1)$$

where,

e = the residual of the problem [-]

A_p = the pulse irradiation coefficient matrix [s^{-1}]

b_p = the pulse irradiation delayed neutron count rate data [$n \cdot s^{-1}$]

A_s = the saturation irradiation coefficient matrix [s^{-1}]

b_s = the saturation irradiation delayed neutron count rate data [$n \cdot s^{-1}$]

x = the solution vector containing group yields [-].

This formulation of the residual removes the need to linearize the count rate, as the residual has the count rate in the denominator. Additionally, this formulation simultaneously treats the pulse and saturation irradiation data,

generating group parameters which fit both sets of data.

The results of this analysis will be used to evaluate which method can provide a solution with the least computational cost while also generating accurate group parameters. The group parameters will be evaluated by calculating the ℓ^2 -norm of the residual for the least squares problem using the generated DNP group parameters over all times for both pulse and saturation irradiations.

5.3 Uncertainty Quantification

Several sources of uncertainty can be considered, including half-lives, emission probabilities, cross-sections, and fission yields. These uncertainties propagate into the DNP group and kinetics parameters [128]. Uncertainties also arise from MSR physical phenomena, including flow rate and reprocessing rates, where the reprocessing rate is correlated with the flow rate. Section 4.3 discusses the sensitivity of the DNP group parameters to changes in MSR phenomena, demonstrating the potential impact of the uncertainties in these values.

These uncertainties are important because they propagate to the DNP group parameters, which affect the reactor's response to different transients. It is important to understand this for safety, security, and safeguards design, as the design should account for these uncertainties.

5.3.1 Data Uncertainties

Uncertainties in the data affect the DNP group parameters through the delayed neutron count rate. The uncertainty in the delayed neutron count rate includes uncertainties in half-lives, emission probabilities, and concentrations. Half-life and emission probability uncertainties are provided by experimental data. The uncertainty in concentrations depends on the method used to generate them. In general, concentration uncertainty arises from fission yields (cumulative or independent, depending on the approach), cross-sections, half-lives, and the neutron flux.

Most uncertainties will be collected from ENDF/B-VIII.0, while the flux uncertainty for the 0D flow model will come from OpenMC. Since Moltres uses a deterministic method to calculate the neutron flux, the flux uncertainty could be calculated using the STOCHASTIC TOOLS module in MOOSE. The STOCHASTIC TOOLS module allows adjustment of other uncertain parameters to calculate the flux uncertainty due to these factors. However, a simpler approach is to use the flux calculated from OpenMC along with its uncertainty. The uncertainties in the data and flux can then be propagated through the Python scripts using the Uncertainties package and applied to various analyses [45].

5.3.2 MSR Uncertainties

In addition to data uncertainties, uncertainties exist in MSR phenomena. These include uncertainties in the in-core residence time, ex-core residence time, and chemical removal rates. Section 4.3 discusses the sensitivity of the DNP group parameters to these terms. It is important to understand the potential uncertainty in these terms, as the residence times and chemical removal rates are not without uncertainty.

The residence time correlates linearly with the flow rate, which is dependent on the temperature. Although the flow rate would also have uncertainty based on variance in the pump, this uncertainty is considered outside the scope of this work. The temperature affects fuel salt density, buoyant force, and viscosity, all influencing the flow rate and residence time. The temperature is influenced by the fission rate, which has uncertainties from the flux and fission cross section. For the 0D flow model, the residence time uncertainties can be conservatively approximated by treating the temperature uncertainty as the difference between the inlet and outlet temperatures

for the core. Assuming a constant mass flow rate, the density change due to the large temperature uncertainty provides a velocity uncertainty, which determines the residence time uncertainty. If this conservative approach to temperature uncertainty results in negligible residence time uncertainty, it can be neglected for the spatially resolved flow model. Otherwise, the spatially resolved flow model can account for this uncertainty using the STOCHASTIC TOOLS module to directly determine residence time uncertainties.

The chemical removal uncertainty is challenging to treat, as there are different chemical processes for removing elements, such as sparging the fuel salt in the core, filtering elements with a nickel filter, and removing elements via liquid-liquid reductive extraction [142]. The removal efficiency of xenon in a gas separation system is given by [118, 142] as:

$$\epsilon = \frac{1 - e^{-K_L a A_C L \left(\frac{1}{Q_{salt}} + \frac{RT}{H} \frac{1}{Q_{He}} \right)}}{1 + \frac{RT}{H} \frac{Q_{salt}}{Q_{He}}}, \quad (5.2)$$

where,

R = the universal gas constant [$L \cdot Pa \cdot mol^{-1} \cdot K^{-1}$]

T = the salt temperature [K]

H = Henry's law constant for solute gas [$Pa \cdot mol^{-1} \cdot L$]

Q_{salt} = the volumetric salt flow rate [$m^3 \cdot s^{-1}$]

Q_{He} = the volumetric helium flow rate [$m^3 \cdot s^{-1}$]

K_L = the liquid phase mass transfer coefficient [$m \cdot s^{-1}$]

a = the gas-liquid interfacial area per unit volume [m^{-1}]

A_C = the contactor cross-section [m^2]

L = the contractor length [m].

The reductive liquid-liquid bismuth extraction distribution coefficient and equilibrium constant depend on the element, temperature, and concentration [108]. In general, an increase in the temperature leads to an increased distribution coefficient and a decreased equilibrium constant. The distribution coefficient represents the ratio of the bismuth phase concentration to the salt phase concentration, and can be thought of as an efficiency term that is multiplied with the mass flow rate to determine the removal rate. However, the distribution coefficient changes with temperature and flow rate differently for each element and fuel salt used.

Propagating the uncertainty in the chemical removal of elements from the fuel salt is challenging, as there are many variables to consider. These include the fuel salt composition, flow rate, redox potential, hydrostatic pressure, interfacial tension, and temperature [146, 157]. Thus, generic equations are not formulated for each possible combination of variables. Instead, only the chemical removal of noble gases and noble metals from the MSRE are investigated [95]. The noble gases are actively pumped out via the off-gas system while noble metals, including molybdenum, technetium, ruthenium, rhodium, palladium, silver, and antimony plate out via reduction by UF_6 . Some elements, specifically selenium, niobium, and tellurium, vary in their behavior. These elements only sometimes plates out based on the redox potential of the salt; a well reduced salt causes them to plate out, otherwise they act as salt seekers [75, 95].

The removal rate, λ_r , of noble gases from the pump bowl via the off-gas system is [74, 95]:

$$\lambda_r = \frac{\dot{V}}{V} \epsilon, \quad (5.3)$$

where,

$\dot{V}$ = the salt flow rate through the gas-liquid separator loop [$m^3 \cdot s^{-1}$]

V = the total volume of salt in the primary loop [m^3]

ϵ = the removal efficiency [-].

For the MSRE, the values are a salt flow rate of 0.003153, a total salt volume of 1.996, and a removal efficiency of 0.026 [134]. For the 0D flow model, this removal occurs while the sample is in the ex-core region and stops while the sample is in the in-core region. This is because the removal occurs in the pump bowl and not inside the core. For the spatially resolved flow model, this removal also occurs in the ex-core region, treated in the same way as the 0D flow model.

The removal rate, λ_r , of noble metals via plate-out is [95]:

$$\lambda_{i,l,r} = \frac{h_{i,l} A_l}{V}, \quad (5.4)$$

where,

$h_{i,l}$ = the mass transfer rate of element i to waste stream l [$m \cdot s^{-1}$]

A_l = the surface area of waste stream l [m^2].

The removal of noble metals does not occur in a single region, unlike the noble gas removal. Instead, there is a distribution of noble metals: 50% are in the hastalloy-N surfaces in the loop, 49% are in the ex-core region, and 1% are in the graphite surfaces in the in-core region [75]. Because 99% of the plate-out occurs in the ex-core region, the removal can be treated as occurring only in the ex-core region. The surface areas and mass transfer rates for the noble metals are given in Table 5.3, recreated from [95].

Component	$h_{i,l} [m \cdot s^{-1}]$	$A_l [m^2]$	$\lambda_{i,l,r} [s^{-1}]$
HX	4.657×10^{-5}	2.93×10^1	6.827×10^{-4}
Fuel loop piping, core and pump volute	1.041×10^{-4}	6.60×10^0	3.441×10^{-4}
Core wall cooling annulus	4.318×10^{-5}	1.43×10^1	3.095×10^{-4}
Core graphite (exposed to salt)	5.334×10^{-6}	1.36×10^2	3.637×10^{-4}
Miscellaneous	-	-	1.695×10^{-4}
Bubbles	4.233×10^{-4}	3.205×10^1	6.7978×10^{-3}
Total	-	-	8.6674×10^{-3}

Table 5.3: Noble metal mass transfer rates, surface areas, and resulting removal rates (recreated from [95] with data from [75]).

5.4 Group Spectra Generation

The delayed neutron energy spectra, χ_d is important for accurate prediction of reactor kinetics [35]. This can be seen in the equation for the effective delayed neutron fraction in Equation (2.25), recreated here as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}.$$

Because the DNP group yields and half-lives are affected by the MSR physical phenomena, it follows that the DNP group spectra are also affected.

The methodology to evaluate the delayed neutron spectra is discussed in Section 2.1.4. The DNP group spectra will be generated using the half-life grouping and fractional grouping methods. The DNP group spectra will then be compared with the spectra generated from a stationary problem with no chemical removal to determine the effect of MSR phenomena on the spectra.

5.5 Spatially Resolved Model Development

Although the 0D flow model provides useful insights, it does not provide a high level of fidelity for any specific geometry. A spatially resolved model will be developed to evaluate the accuracy of the 0D flow model. This model verifies whether the 0D flow model produces sufficiently accurate DNP group parameters. Section 3.2.2 discusses the spatially resolved model in detail. This spatially resolved model will compare multiple levels of fidelity for generating DNP group parameters: 0D flow model using cumulative fission yields, 0D flow model with depletion, spatially resolved flow model using cumulative fission yields, and spatially resolved flow model with depletion. Each step increases computational cost, so if a simpler method generates DNP group parameters with negligible differences, it will be used preferentially.

The process for comparing the 0D flow model and the spatially resolved model is as follows: the spatially resolved model runs and generates DNP group parameters and residence times, the residence times are used to feed into the 0D flow model, the resulting DNP group parameters are compared between the two models. If the results are within the uncertainty bounds of each other, then this indicates that the 0D flow model only needs residence time data for a given MSR to generate the DNP group parameters. If they do not match, then a spatially resolved approach will be necessary to properly generate the MSR DNP group parameters.

5.6 Kinetics Simulations

Kinetics simulations are critical for validating the DNP group parameters using MSRE transients and for investigating transients affecting security, safety, and safeguards design decisions. Section 3.5 discusses the proposed simulations in detail. For safety design, the ULOF and UTOP are investigated for various reactor parameters to assess the impact of the updated DNP group parameters. For security design, sabotage-initiated transients are assumed to build off those studied for safety design. For safeguards design, comparisons are made between diversion scenarios with different PR times and material quantities removed. These diversion scenarios alter the fuel composition, affecting the effective delayed neutron fraction and reactor behavior during transients [166]. The updated DNP group parameters will be used to investigate the SNM removal which could remain undetected while using previous DNP group parameters.

The kinetics simulations will use Moltres and the MSRE to demonstrate how the DNP group parameters affect reactivity and temperature responses in various scenarios. These results can be compared with other analyses from the literature [63, 152]. The results will identify which transients are more sensitive to the updated data and highlight potential concerns. A PRK model can also be used for simple reactivity insertion tests with uncertainty tracking to more straightforwardly demonstrate differences between the DNP group parameters.

5.7 Tool Development

The final proposed extension to this work is the release of an open-source Python package. This package will provide users with the capability to generate DNP group parameters for any MSR, with any fissile nuclide, energy spectra, and desired number of DNP groups. The plan is to offer a customizable and modular tool which is easily accessible, enabling users to study more effects of these updated DNP group parameters.

There are many additional effects to study which would be of interest, including: more refined data generation, investigation of other fissile nuclides and energy spectra, further optimizations to the methodology, and inclusion of additional MSR physical phenomena such as thermochemistry. Other future work of interest would be experimental results replicating the methodology proposed in this work, specifically, irradiation of a fissile sample in a circulating loop and counting of the delayed neutrons from that sample. Work in this area will become streamlined with the development of this tool, while also allowing users to modify the tool to suit any other potential needs.